Zero-Shot Cross-City Trajectory Prediction Using Hypernetworks

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***Abstract*—City-wide mobility prediction models typically rely on either training with extensive local trajectory data or applying transfer learning from data-rich cities to those with limited data. In both cases, the resulting models are specialized to a specific target city for which they require at least some trajectory data to adapt. Consequently, they cannot generalize to cities unseen during training and are inapplicable where no mobility data exist. In this work, we propose H0xtra, a novel approach that enables zero-shot trajectory prediction in entirely unseen cities. H0xtra leverages a hypernetwork to generate city-specific location embeddings from spatial distributions of points of interest, e.g., restaurants or stores. These embeddings capture city-agnostic location semantics, enabling a transformer to learn universal trajectory patterns across cities. At inference, H0xtra performs zero-shot transfer without requiring any mobility data or retraining. Adaptation requires only points of interest data, which are often publicly available, to generate location embeddings specific to the target city. Trained only on a small set of source cities, H0xtra achieves strong zero-shot generalization. In our experiments, the zero-shot performance achieves an accuracy improvement of 11.3% and an average displacement error reduction of 11.5% on average compared to state-of-the-art non-zero-shot baselines. Our code can be accessed at https://github.com/DLR-Prot-of-Terrestrial-Infrastructures/H0xtra.**

***Index Terms*—Zero-Shot, Hypernetwork, Mobility Prediction, Cross-City Mobility**

I. INTRODUCTION

Detailed information on human mobility is crucial for improving traffic management, obtaining economic insights, and controlling epidemic spread [1]. Therefore, the ability to predict realistic human trajectories is of great value. Existing models typically accomplish this task through training on large amounts of trajectory data from a single city [2]. However, for many cities, such data may be unavailable or restricted by privacy regulations, hindering the applicability of such models.

Recently, several approaches have been proposed to transfer knowledge from data-rich source cities to target cities with limited mobility data [3]–[6]. These approaches typically rely on one of two strategies: either a two-stage training that starts with pre-training on source-cities followed by fine-tuning on target city data [6], or domain adaptation techniques that align feature distributions between source and target cities [4], [7]. A significant limitation of both strategies is their inevitable dependence on target city trajectory data for adaptation during training. While such approaches have been demonstrated to enhance performance in few-shot settings with limited data availability, they remain inherently inapplicable to cities without any trajectory data.

Settings without mobility data available for a target city necessitate models to generalize to completely unseen urban environments. However, different cities exhibit different spatial characteristics, resulting in a *domain shift* between trajectories from different cities [8]. Consequently, the *zero-shot transfer* from source cities to a target city requires the model to overcome this domain shift, which poses a significant challenge [9]. For a successful zero-shot transfer, a model must be able to infer key characteristics of the unseen target city, such as its spatial layout or the semantics of different locations, and relate these characteristics to the previously seen source cities. We argue that this level of generalization can effectively be achieved by learning a joint representation space that captures cross-city regularities and unifies trajectory representations. Such a representation space enables the model to extrapolate learned regularities to unseen cities, ultimately supporting zero-shot trajectory prediction.

To meet these requirements, we propose a novel model architecture that comprises a **h**ypernetwork for **zero**-shot **cross**-city **tra**jectory prediction, named **H0xtra**. At its core, H0xtra leverages the hypernetwork [10] to learn a trajectory representation space that decouples trajectories from their city of origin. This disentanglement allows the model to be trained across multiple cities and to generalize to new ones.

The architecture consists of two components that are trained in parallel: (i) a CNN U-Net [11], which serves as the hypernetwork, generating parameters for (ii) a Transformer [12]. The U-Net transforms discrete spatial distributions of points of interest (POIs) into city-specific location embeddings that contextualize each location with the entire city. As a result, the embeddings encode relative urban semantics rather than absolute, isolated location attributes. These embeddings parameterize the Transformer's input and output layers, defining a representation space in which the Transformer can model trajectories in a city-agnostic manner.

This architecture enables H0xtra to overcome the domain shift between cities with varying spatial and structural properties: By training jointly on multiple cities, the U-Net can learn to associate POI distributions with trajectories in a generalized way, producing embeddings that are transferable across cities. At inference, the model adapts to an unseen city using its POI data only, which are typically publicly available. By generating

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corresponding location embeddings, trajectories in the target city are projected to the previously learned representation space. This *zero-shot transfer* permits the Transformer to perform next-location prediction or generate full synthetic trajectories without requiring any retraining or mobility data from the target city.

Our contributions can be summarized as follows.

- We propose H0xtra, a novel architecture that leverages a *hypernetwork* to learn city-agnostic trajectory representations for zero-shot cross-city mobility modeling. Contrary to existing approaches, H0xtra enables zero-shot transfer to unseen target cities only using widely available POI data, without requiring any retraining.
- With H0xtra, we introduce a new paradigm for multi-city trajectory modeling that complements existing transfer learning approaches by directly unifying the trajectory representation space across multiple cities.
- We conduct extensive evaluations on real-world datasets, demonstrating that H0xtra achieves strong generalization while maintaining high accuracy: in many cases, the zero-shot performance even surpasses state-of-the-art non-zero-shot baselines.

The rest of the paper is organized as follows. Section II briefly discusses related work. Section III introduces preliminaries and formalizes the task of zero-shot cross-city trajectory prediction. In Section IV, we present the architecture of H0xtra. The experiments and results are presented in Section V before the paper concludes with Section VI.

II. Related Work

This section presents previous work related to our approach. We first discuss recent cross-city mobility modeling approaches, followed by an overview of research on hypernetworks for domain generalization.

A. Cross-City Mobility Modeling

Cross-city mobility models aim to leverage rich data from source cities to improve performance in data-scarce or data-absent target cities. Existing approaches predominately transfer learned representations either through pre-training on source city data followed by few-shot fine-tuning on the target city [6], [13], or by aligning feature distributions among source and target cities [4], [7], [14]–[17].

However, only a fraction of such methods are capable of zero-shot predictions in cities entirely lacking mobility data. Rong et al. [14] proposed a domain-adversarial framework that aligns structural features between a source and a target city to transfer origin-destination crowd flows. He et al. [18] introduce a multi-stage pipeline, involving feature alignment of source cities, origin-destination transfer to the target city, and path generation.

In contrast, our approach unifies trajectories from multiple cities through end-to-end training, enabling zero-shot prediction in any unseen city without the need for retraining or explicit city-to-city alignment.

Recently, the generalization capabilities of *large language models* have been increasingly explored for urban mobility modeling [19]–[24]. Several of these approaches support zero-shot application in unseen cities, e.g., for next-location prediction [22] and crowd flow modeling [23], [25]. However, they often require extensive data preprocessing and entail high computational cost, whereas our method is directly transferable to unseen cities and remains lightweight during inference.

B. Hypernetworks for Domain Generalization

Hypernetworks are neural networks that generate parameters of another network [10]. They expand modeling capacity and enable models to dynamically adapt to varying contexts. This flexibility has been explored for domain generalization in fields such as multi-task language modeling [26]–[31], and image classification [32], translation [33]–[35] and generation [36].

Typically, such models adapt a subset of parameters based on an example of the target domain, provided either as the task input (e.g., a source image) or as an auxiliary signal that indicates context.

A particularly related approach is that of Minixhofer et al. [37], who used a hypernetwork to adapt a language model to unseen tokenizers. Similarly to their work, our approach does not require explicit examples from the target domain, but instead uses city representations as domain abstractions.

III. Task Formalization

In this section, we formally define the problem of zero-shot cross-city trajectory prediction.

A. Trajectory Modeling

Let $\mathcal{C}$ denote a city represented by a set of n discrete locations $\mathcal{L}$ that are spatially arranged in a grid of size $h \times w$. A trajectory within city $\mathcal{C}$ is defined as a sequence $\tau = [(l_1, t_1), (l_2, t_2), \ldots, (l_m, t_m)]$ of pairs (l_i, t_i), consisting of a location $l_i \in \mathcal{L}$ visited at timestamp t_i. A set of k trajectories within city $\mathcal{C}$ is given by $\mathcal{T} = \{\tau_j, j = 1, \ldots, k\}$. Additionally, let

$$m \in \mathbb{R}^{h \times w \times \text{num_poi}} \tag{1}$$

represent the structure of city $\mathcal{C}$ as the relative occurrences of different POIs (e.g., restaurants) within each location.

In this work, we consider two core tasks in trajectory modeling: *next-location prediction* and *trajectory generation*, which we formalize as follows.

Next-location prediction is defined as inferring the future location of a given trajectory. Formally, for a given trajectory τ, this future location is sampled from a distribution over $\mathcal{L}$, given as

$$l_{m+1} \sim P(l \,|\, \tau), \quad l_{m+1} \in \mathcal{L}. \tag{2}$$

For each possible location, this distribution represents the visitation probability, given the known past trajectory.

Trajectory generation is defined as creating a trajectory without past context locations. For the trajectory, we assume that the probability of visiting a location at time t_i is independent of the locations that are visited at later timestamps t_k with $k > i$. Consequently, trajectories may be

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generated by autoregressively sampling from the distribution introduced in (2). The overall probability of trajectory $\tau = [(l_1, t_1), (l_2, t_2), \ldots, (l_m, t_m)]$ is given by

$$P(\tau) = P(l_1) \cdot \prod_{i=2}^{m} P(l_i \,|\, \tau_{:i-1}), \quad l_1, l_i \in \mathcal{L},$$

where, $\tau_{:i}$ denotes the sub-trajectory up to index i. Here, a trajectory is represented as a sequence of locations only, with corresponding timestamps given implicitly. Consequently, this definition assumes a continuous location sequence without temporal gaps.

B. Cross-City Trajectory Prediction

Let $\mathcal{T}_\mathcal{C}$ be the set of observed trajectories in city $\mathcal{C}$. We approximate the distribution in (2) using a parametric model $f(\cdot, \theta)$ trained on $\mathcal{T}_\mathcal{C}$. The next location in trajectory τ can then be sampled as

$$l \sim f(\tau, \theta_{\mathcal{T}_\mathcal{C}}).$$

To support multi-city mobility modeling, we extend the model with city-specific representations $m_\mathcal{C}$. Let $\mathbf{C} = \{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_b\}$ be a set of source cities, each with observed trajectory data $\mathbf{D} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_b\}$ and city structure representations $\mathbf{m} = \{m_1, m_2, \ldots, m_b\}$. We train a multi-city model $f(\cdot, \cdot, \theta_\mathbf{C})$ with parameters $\theta_\mathbf{C} = \theta_{(\mathbf{D},\mathbf{m})}$ jointly on the trajectory data $\mathbf{D}$ and their corresponding city structures $\mathbf{m}$.

For an arbitrary target city $\hat{\mathcal{C}}$, whether seen during training ($\hat{\mathcal{C}} \in \mathbf{C}$) or not ($\hat{\mathcal{C}} \notin \mathbf{C}$), we can sample the next location in trajectory τ as

$$l \sim f(\tau, m_{\hat{\mathcal{C}}}, \theta_\mathbf{C}). \tag{3}$$

The zero-shot setting refers to the case $\hat{\mathcal{C}} \notin \mathbf{C}$, in which the model has not accessed any trajectory data from the target city during training. Instead, it relies solely on characteristics learned from source city trajectories combined with their corresponding city structures. A representation of the target city alone then suffices to enable generalization, allowing both *next-location prediction* and full *trajectory generation* in previously unseen environments.

In this work, we propose an architecture for the parametric model specified in (3). This architecture enables unifying trajectory representations by learning generalized location embeddings using city representations $\mathbf{m}$. The next section provides architectural details of our model.

IV. Model Architecture

To enable zero-shot generalization to unseen cities, H0xtra relates trajectory patterns to structural characteristics of the urban environment. As shown in Figure 1, H0xtra consists of two components: (i) a CNN U-Net as *city structure encoder* that generates location embeddings from spatial data, and (ii) a *Transformer* as trajectory predictor that captures sequential dependencies. The city structure encoder acts as a hypernetwork, i.e., a neural network that generates parameters for another network [10]. In our architecture, it injects city-specific weights into the Transformer to unify trajectory representations. This unified representation space enables generalization to previously unseen target cities.

A. City Structure Encoder

As defined in (1), a city is represented by POI distributions m, which we augment with 2D sinusoidal positional encodings [12] to inject spatial proximity between locations. From these representations, the city structure encoder generates location embeddings. To enable zero-shot generalization, these embeddings must jointly reflect a location's semantics as well as its relation to the entire city. This requirement motivates our choice of a CNN U-net as city structure encoder [11]. The U-Net fuses coarse, city-wide information with fine-grained, location-specific data by progressively compressing and expanding the city structure. This contextualizes each location with global, city-level features, producing embeddings that are locally homogeneous yet globally discriminative.

Let $\mathcal{U}(\cdot)$ denote the U-Net. It consists of L downsampling CNN layers followed by L upsampling CNN layers:

$$\begin{aligned} D_l &= d_l(D_{l-1}), \\ U_l &= u_l(U_{l-1}, D_{L-l}), \end{aligned} \qquad (l = 1, \ldots, L)$$

where D_0 is the input to the downsampling block, $U_0 = D_L$ the initial input to the upsampling block, and U_L the final output. The functions $d_l(\cdot)$ and $u_l(\cdot, \cdot)$ denote the down- and up-projecting CNN layers, implemented as

$$\begin{aligned} d_l(x) &= \text{ReLU}(\text{conv}(\text{ReLU}(\text{conv}(\text{maxpool}(x))))), \\ u_l(x, y) &= \text{ReLU}(\text{conv}(\text{ReLU}(\text{conv}(\text{upconv}(x) + y)))). \end{aligned}$$

Here, the kernel size for maxpool is set to 2×2.

The overall hypernetwork consists of an initial convolutional layer, followed by the U-Net, and a final convolution and normalization. The output is reshaped to represent a matrix of num_locs $= h \cdot w$ location embeddings of dimension embed_dim_loc:

$$\mathcal{H}(m) = \text{LN}(\text{conv}(\mathcal{U}(\text{conv}(m))) \in \mathbb{R}^{\text{num_locs} \times \text{embed_dim_loc}}.$$

B. Transformer as Trajectory Predictor

As sequence model in our architecture we use a decoder-only Transformer with masked multi-head self-attention for causal next-location modeling [12]. Its role is to capture sequential characteristics of human trajectories.

For each trajectory, timestamps and location IDs are embedded separately and then concatenated to a joint embedded sequence. Formally, the embedding layers are defined as:

$$\begin{aligned} f_{\text{tim}}(t) &= \mathbf{M}^T t, \\ f_{\text{loc}}(l) &= \mathbf{W}^T l, \end{aligned}$$

with embedding matrices $\mathbf{M} \in \mathbb{R}^{\text{num_times} \times \text{embed_dim_tim}}$ and $\mathbf{W} \in \mathbb{R}^{\text{num_locs} \times \text{embed_dim_loc}}$, and one-hot vectors $t \in \{0,1\}^{\text{num_times}}$ and $l \in \{0,1\}^{\text{num_locs}}$, representing timestamps and location, respectively.

The embedded sequence is passed through the Transformer layers, which capture dependencies across time steps and infer

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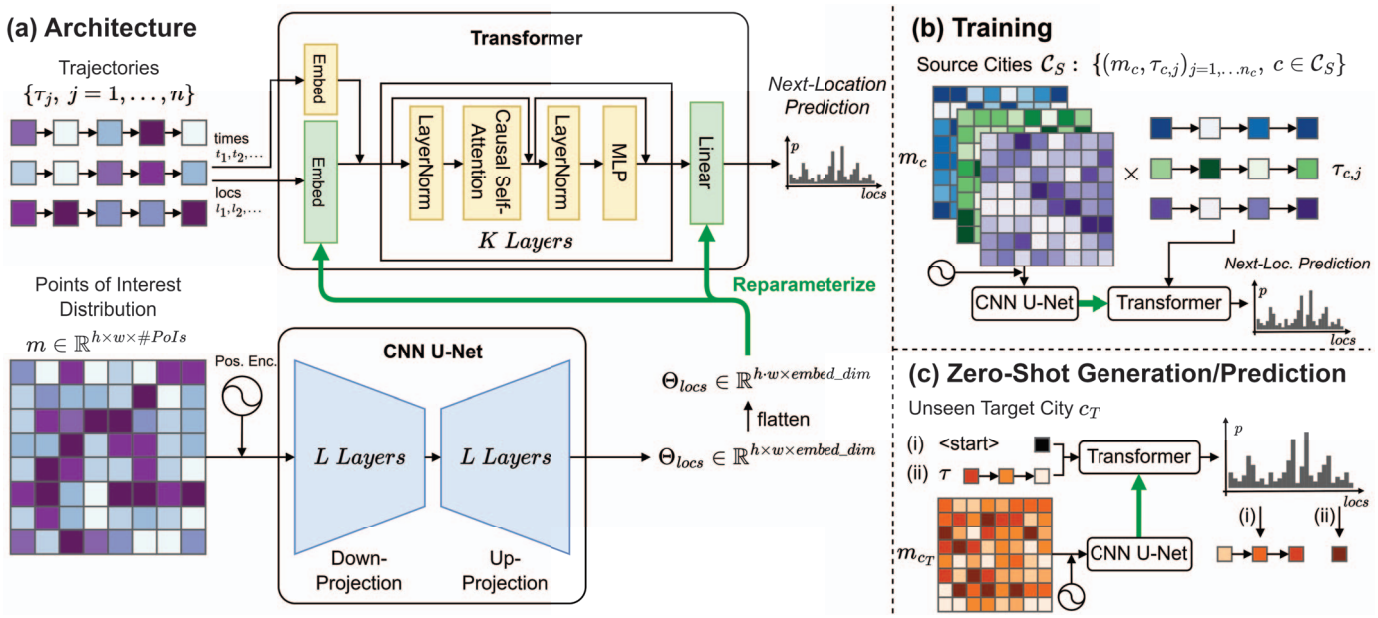


Fig. 1. Overview of the proposed zero-shot modeling framework. **(a) Architecture:** A CNN U-Net generates location embeddings from spatially arranged POI distributions. These location embeddings parameterize a decoder-only Transformer. **(b) Training:** The model is trained on trajectories from a set of source cities $\mathbf{C}$, learning sequential patterns in human trajectories that generalize across urban environments. **(c) Zero-Shot Application:** For inference for an unseen target city, the U-Net produces location embedding specific to that target city using only POI data. These embeddings adapt the Transformer to the new city, enabling *zero-shot next-location prediction* and *zero-shot trajectory generation*.

a representation of the candidate for the next location of any coherent subsequence. These representations serve as input to a linear layer projecting to the dimension of the location embeddings, followed by final linear layer that computes logits over all possible locations:

$$f_{\text{proj}}(x) = \mathbf{N}x,$$
$$f_{\text{out}}(x) = \mathbf{W}x,$$

where $\mathbf{N} \in \mathbb{R}^{\text{embed_dim_loc} \times \text{embed_dim_tim+embed_dim_loc}}$, the weight matrix $\mathbf{W}$ is shared with the location embedding layer, following the weight tying principle [38]. With weight tying, the Transformer operates on one joint embedding space for input embedding and logits computation.

A final softmax function projects the logits to a discrete probability distribution that approximates the distribution defined in (2). This distribution is used for sampling future locations of trajectories.

C. Cross-City Generalization via Hypernetwork Embeddings

As illustrated in Figure 1, the city structure encoder functions as a hypernetwork, parameterizing both the location embedding and final linear layers of the Transformer:

$$f_{\text{loc}}(l) = \mathcal{H}(m)^T l,$$
$$f_{\text{out}}(x) = \mathcal{H}(m)x.$$

Through the learned location embeddings, the Transformer can implicitly capture dependencies between semantic roles and spatial proximity of locations within trajectories. In parallel, the city structure encoder learns to embed how POI distributions and city-wide layout influences location visitation frequencies. This interplay enables city-agnostic trajectory modeling.

The overall model can be directly trained on pairs of trajectory and corresponding city structure data of an arbitrary number of cities. By training on multiple cities, the hypernetwork learns to generate location embeddings that unify trajectory representations across diverse urban environments.

While the hypernetwork increases training cost, inference remains efficient. Zero-shot transfer to unseen cities only requires a city structure (cf. (1)) that may be derived from publicly available POI data. A single forward pass through the hypernetwork updates the parameters of the embedding and final linear layers once, adapting the Transformer into a target city-specific trajectory model.

V. Experiments

We design the experiments to explicitly analyze H0xtra's zero-shot performance in a previously unseen city. The evaluation encompasses two different tasks, *next-location prediction* and *trajectory generation*, for which we train H0xtra on all available cities except a chosen target city. For both tasks, we compare the performance against a range of non-zero-shot baselines.

A. Setup

1) Data: We conduct experiments with data from four cities, denoted as city A, B, C, and D. The data is provided by LY Japan and publicly available [40], [41]. The cities are discretized into 200 × 200 grids of 500m × 500m cells. For each city, the data contains trajectories over a period of 75

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TABLE I

PERFORMANCE COMPARISON FOR **ZERO-SHOT CROSS-CITY** NEXT-LOCATION PREDICTION. FOR EACH METRIC, THE DATA IS AVERAGED OVER ALL FOUR CITIES. ALL BASELINES ARE TRAINED AND TESTED ON THE SAME CITY. CONTRASTING THE BASELINES, H0XTRA WAS TESTED EXCLUSIVELY ON CITIES **NOT** INCLUDED IN THE TRAINING DATA. THE PROVIDED RELATIVE DEVIATIONS REFER TO THE BASELINE WITH THE BEST PERFORMANCE.

	Metric	@1 (↑)	@5 (↑)	@10 (↑)	@20 (↑)	MRR (↑)	ADE (↓)
non-zero-shot	HA	0.063	-	-	-	-	4759.2
	RNN	0.146	0.318	0.388	0.456	0.229	9349.4
	LSTM	0.136	0.287	0.348	0.410	0.210	10 777.1
	DeepMove [39]	0.216	0.442	0.522	0.596	0.322	5459.0
	Transformer	0.257	0.546	0.640	**0.714**	0.389	2547.3
	COLA [6]	0.247	0.530	0.627	0.706	0.376	2455.9
	Zero-Shot Prediction (H0xtra)	**0.286** +11.3%	**0.569** +4.2%	**0.645** +0.8%	0.707 -1.0%	**0.416** +6.9%	**2174.4** –11.5%

days for 100k, 25k, 20k, and 6k individuals, respectively, as well as POI occurrences for each cell. The trajectories are discretized in 30-minute intervals, i.e., contain maximum 48 location reports per day.

We follow previous work and select individuals with frequent visitation reports [6]. For *next-location prediction*, these are individuals with at least five reports per day across the entire observation period, yielding 11882 2225, 1431, and 1490 individuals for city A, B, C, and D, respectively. For *trajectory generation*, which requires single days, we include individuals with at least five reports on any active day, resulting in 17005, 3319, 2067, 2075 individuals, respectively. Each set is split into training, validation, and test partitions with a ratio of 7:1:2.

We further divide the training trajectories for *next-location prediction* into subtrajectories of approximately equal length ($\sim$250 locations) to increase sample size. For *trajectory generation*, we follow previous work [6], [42] and split the trajectories into reports of single days, which we forward fill to close potential temporal gaps, resulting in full one-day trajectories.

2) Baselines: Since zero-shot cross-city mobility modeling has attained little attention so far, there is a lack of established baselines for comparison. Therefore, we assess our model against arguably the most challenging baselines, namely models that were trained *and* tested on the same city, referred to as *non-zero-shot models*.

We choose two types of baselines: (i) *single-city models*, trained and tested on the same single city, and (ii) *multi-city models*, trained on all cities and tested on a specific target city. For the single-city models, we include the historical average[1] (HA), RNN, LSTM [43], DeepMove[1] [39], MoveSim[1] [42], and a Transformer [12]. We select COLA [6] as a recently proposed multi-city model that demonstrates promising generalization across multiple cities. For all baselines, we adopt the configurations reported in the corresponding papers, where available. For the Transformer baseline, we choose the exact same configuration as for the Transformer module in H0xtra.

Furthermore, we included two non-zero-shot versions of our approach: **H0xtra-SC** and **H0xtra-MC** which are single-city and multi-city versions of our approach, respectively.

[1]Due to their design, we use HA and DeepMove only for *next-location prediction* and MoveSim only for *trajectory generation*.

3) H0xtra Training Configuration: We use 4 Transformer layers with 16 attention heads and an embedding dimension of 32, and 8 U-Net layers with a filter size of 8×8. The training is performed with cross-entropy loss and cosine learning rate decay, starting at a learning rate of 5e-3. For *next-location prediction*, we use a batch size of 16 and train for 20 epochs; for *trajectory generation*, we use a batch size of 128 and train for 10 epochs. We select the final models by validation loss on the *source cities*, thereby preserving the zero-shot setting. All trainings were done on an Nvidia A100 GPU using PyTorch 2.6. With this setup, a single training step took about 0.2 seconds for next-location prediction and about 0.9 seconds for trajectory generation.

4) Metrics: For *trajectory generation*, we follow prior work [6], [42] and reduce both simulated and ground truth trajectories to statistical distributions, which we compare for similarity by *Jensen-Shannon divergence* (JSD). The statistics include (i) *Distance:* the spatial gap between consecutive locations, (ii) *Radius:* the average distance to the trajectory center, (iii) *Duration:* the time spent at each stay in one location, (iv) *Periodicity:* the number of distinct visited locations, (v) *I-rank:* the visitation frequencies of the 20 most visited locations per individual, and (vi) *G-rank:* the visitation frequencies of the 1000 most visited locations overall.

For *next-location prediction*, we evaluate performance by *top-k accuracy* (@k), measuring whether the correct location is among the top k candidates of highest predicted probability, and *mean reciprocal rank* (MRR), reflecting the rank position of the correct location. In addition, we report the *average displacement error* (ADE) in meters between the predicted and correct location.

5) Training Setting: For both tasks, we trained all models with causal next-location prediction as objective. For the task of *next-location prediction*, we evaluate the models on their ability to predict the next step given a context trajectory of length 40. For *trajectory generation*, training includes empty location and time stamp tokens as start tokens, which serve as starting element to autoregressively generate new full-day trajectories during testing.

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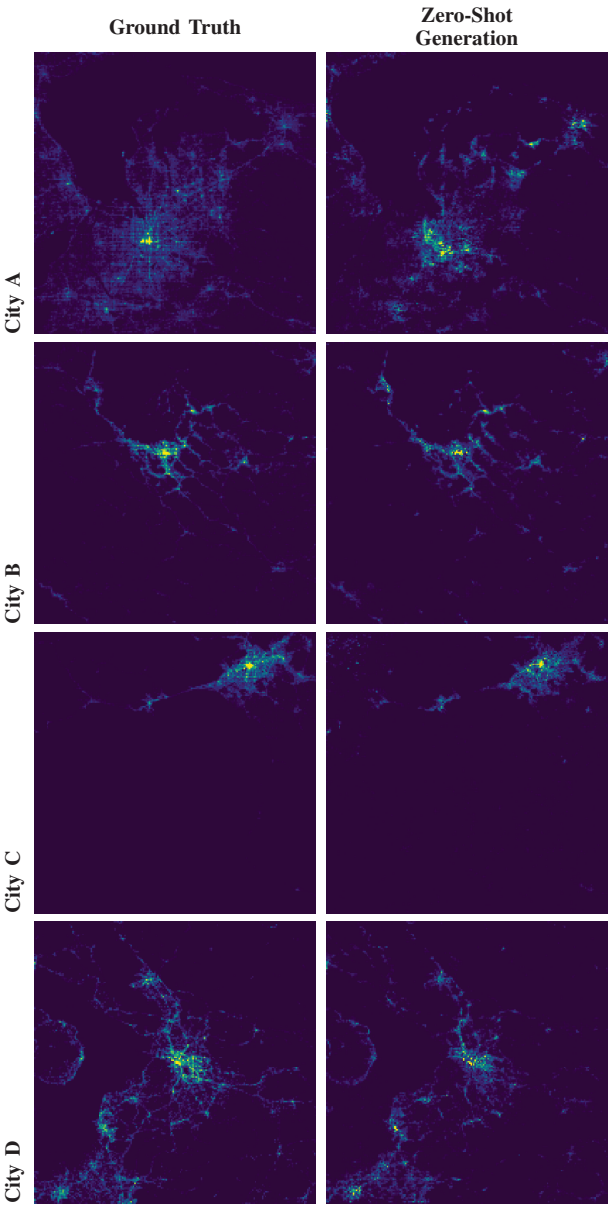


Fig. 2. Aggregated population density from real and generated trajectories. Lighter colors refer to higher population volumes. To improve contrast, values above the 99.7th percentile were clipped.

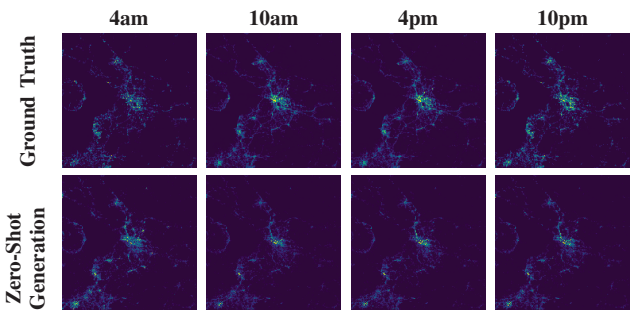


Fig. 3. Daily course of population densities from real and generated data. We chose city D as representative city that is spread across a wide area but has a distinct city center. Lighter colors refer to higher population volumes. To improve contrast, values above the 99.7th percentile were clipped.

B. Zero-Shot Cross-City Results

This section exclusively evaluates the zero-shot performance of our approach both for *next-location prediction* and *trajectory generation*.

1) Next-Location Prediction: Table I shows that H0xtra achieves on average higher accuracy and greater spatial consistency than non-zero-shot baselines. We observe an enhancement in accuracy (i.e., @1) of at least 11.3% and a decrease in ADE of at least 11.5% across all selected baselines. These results demonstrate that the hypernetwork effectively learns to generate location embeddings that unify trajectories across cities in a shared representation space. The margin by which H0xtra outperforms all baselines underscores the universality of this representation space, indicating that it is more representative than a space exclusively learned with target city data.

2) Trajectory Generation: The results for *trajectory generation* suggest that the learned location embeddings successfully capture the functional roles of locations within a city. As demonstrated in Figure 2, the population densities derived from generated trajectories closely reflect the true spatial characteristics. However, H0xtra tends to overestimate visitations in highly frequented areas (e.g., the centers of city B and D) while underestimating areas with lower visitation frequencies (e.g., for city A).

For city A, we observe a deviation of population densities that is greater than for the other cities. The reason may be the difference in training data volume: cities B, C, and D each have 80% fewer training trajectories than city A. This smaller set of training trajectories may lack heterogeneity and thus impair the model's ability to generalize. Alternatively, the deviation in densities could originate from differences in urban layout. City A is a wide-spread metropolitan area, whereas cities B, C, and D are much more concentrated. A zero-shot transfer to city A may partially transfer the more clustered shape of the source cities, resulting in localized hot-spots rather than a broader population spread.

Figure 3 illustrates the daily temporal evolution of the population density. While the previously mentioned overestimation of central areas remains visible, a general parallelism between generated and true data becomes apparent. At night (4am), the population is dispersed throughout the city area as most individuals stay at home. In the morning (10am), individuals are at their workplace, resulting in increased concentration in the city center, which persists into the afternoon (4pm). In the evening (10pm), the densities begin to diffuse towards a more dispersed state once again.

C. Comprehensive Performance Comparison

So far, we showed H0xtra's superiority for zero-shot *next-location prediction* and its general effectiveness for zero-shot *trajectory generation*. This section extends our previous evaluation by two variations of our approach to further analyze its capabilities for general multi-city trajectory modeling. In

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TABLE II

NEXT-LOCATION PREDICTION PERFORMANCE. EACH BLOCK REPRESENTS A SINGLE CITY THAT (I) FOR THE SINGLE-CITY MODELS REPRESENTS THE TRAINING AND TEST CITY, (II) FOR THE MULTI-CITY MODELS TRAINED ON ALL FOUR CITIES REPRESENTS THE TEST CITY, AND (III) FOR OUR ZERO-SHOT MODEL REPRESENTS THE TEST CITY. BEST RESULTS PER METRIC ARE IN **BOLD** AND SECOND-BEST ARE UNDERLINED. WITH AN ASTERISK*, WE INDICATE WHETHER THE ZERO-SHOT PREDICTION SURPASSES ALL BASELINES.

Metric	@1 (↑)	@5 (↑)	@10 (↑)	@20 (↑)	MRR (↑)	ADE (↓)	@1 (↑)	@5 (↑)	@10 (↑)	@20 (↑)	MRR (↑)	ADE (↓)
	City A						**City B**					
HA	0.068	-	-	-	-	5202.4	0.063	-	-	-	-	5010.9
RNN	0.152	0.361	0.455	0.543	0.251	3924.3	0.170	0.354	0.421	0.486	0.258	9982.6
LSTM	0.134	0.295	0.370	0.450	0.214	6541.4	0.168	0.341	0.406	0.468	0.251	11136.4
DeepMove [39]	0.253	0.510	0.603	0.683	0.372	2793.0	0.230	0.455	0.531	0.607	0.335	5844.8
Transformer	0.284	0.587	0.671	0.738	0.422	2129.1	0.276	0.574	0.668	0.742	0.411	2790.5
COLA [6]	0.271	0.565	0.657	0.729	0.404	2283.9	0.252	0.541	0.640	0.720	0.383	2665.5
H0xtra (ours)	0.257	0.520	0.602	0.670	0.377	2351.7	0.315*	0.612*	0.680*	0.742*	0.453*	2213.6*
H0xtra-SC (ours)	0.315	0.612	0.686	0.745	0.450	2001.1	0.280	0.574	0.668	0.745	0.414	2257.3
H0xtra-MC (ours)	**0.329**	**0.629**	**0.695**	**0.750**	**0.466**	**1982.0**	**0.330**	**0.640**	**0.712**	**0.772**	**0.471**	**2127.7**
	City C						**City D**					
HA	0.063	-	-	-	-	4227.7	0.057	-	-	-	-	4591.8
RNN	0.182	0.371	0.445	0.517	0.274	4942.9	0.078	0.186	0.229	0.279	0.132	18547.8
LSTM	0.171	0.341	0.404	0.466	0.254	5970.8	0.072	0.171	0.211	0.256	0.121	19459.6
DeepMove [39]	0.231	0.481	0.565	0.636	0.347	3776.9	0.149	0.323	0.391	0.456	0.233	9421.5
Transformer	0.259	0.550	0.635	0.706	0.390	2150.5	0.208	0.474	0.585	0.671	0.332	3119.0
COLA [6]	0.261	0.554	0.640	0.716	0.394	2270.9	0.202	0.460	0.570	0.660	0.323	2603.4
H0xtra (ours)	0.297*	0.579*	0.646*	0.703	0.426*	2044.1*	0.273*	0.566*	0.649*	0.714*	0.406*	2088.4*
H0xtra-SC (ours)	0.279	0.580	0.664	0.727	0.416	2023.3	0.219	0.498	0.594	0.682	0.347	2382.9
H0xtra-MC (ours)	**0.314**	**0.616**	**0.682**	**0.734**	**0.454**	**1953.3**	**0.281**	**0.589**	**0.670**	**0.735**	**0.420**	**2062.0**

Color coding: Single-city approaches, multi-city approaches, zero-shot cross-city approach

this analysis, we include the *non-zero-shot single-city* (H0xtra-SC) and *multi-city* (H0xtra-MC) versions of H0xtra, which are trained on a single city and all cities, respectively.

1) Next-Location Prediction: Table II shows that H0xtra-MC consistently outperforms all other models, including the *single-city* version, highlighting the effectiveness of unifying trajectory representations for cross-city trajectory modeling.

Notably, our zero-shot H0xtra even surpasses the non-zero-shot H0xtra-SC in many cases. Given that both models have access to trajectory *and* POI data, this implies that H0xtra has learned to generalize trajectories across cities so effectively that these generalized representations may be more informative for a given target city than its original data. Further, in comparison to the other baselines, this finding confirms that the strong zero-shot performance cannot be attributed to the inclusion of POI data alone, but rather to H0xtra's ability to unify trajectories across cities.

For cities B, C, and D, our three model versions consistently achieve the top three ranks among all models. For city A, however, while H0xtra-MC and H0xtra-SC occupy the top ranks, we observe a zero-shot performance that is surpassed by multiple non-zero-shot baselines. As discussed in Section V-B2, this may result from the lower training data volume for city B, C, and D, or from clear differences in urban layout.

2) Trajectory Generation: Table III shows an overall dominance of H0xtra-SC and H0xtra-MC, particularly in terms of spatial consistency, as evidenced by the best scores for *Distance* and *Radius* across all cities. This spatial consistency also holds for the zero-shot H0xtra, which achieves *Radius* scores close to or surpassing most baselines, demonstrating H0xtra's ability to encode spatial proximity and its influence on individuals' mobility decisions.

For *Duration*, *Periodicity*, and *I-rank*, reflecting temporal characteristics, our models perform strong overall, although some baselines achieve comparable results.

Conversely, the results for *G-rank* show a superiority of MoveSim, RNN, and LSTM, which outperform our models, partially to a considerable extent. The objective of *G-rank* is to evaluate the overall spatial realism of generated trajectories. However, the metric lacks expressiveness when a model drastically overestimates the visitation frequencies of only very few locations, as is the case with MoveSim (see Figure 4).

To ensure a comprehensive evaluation of the overall spatial realism, we further analyze the population densities derived from trajectories, shown in Figure 4.

A key difference between our models and the baselines lies in the spatial cohesiveness of the densities. Our learned location embeddings enable extrapolation of semantics and expected visitation frequencies to unseen locations. Unlike baseline models, which primarily generate trajectories that adhere to locations seen during training, our models can generate trajectories that contain unseen locations while preserving spatial and semantic realism. This results in more cohesive and comprehensive densities, whereas those from baselines partially exhibit scattered locations with unrealistic gaps in between (e.g., for the Transformer and COLA).

D. Analysis of Generalization Capabilities

While the preceding sections demonstrated the overall performance, this section focuses on the components that influence the generalization capabilities of our approach.

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TABLE III

TRAJECTORY GENERATION PERFORMANCE. EACH BLOCK REPRESENTS A SINGLE CITY THAT (I) FOR THE SINGLE-CITY MODELS REPRESENTS THE TRAINING AND TEST CITY, (II) FOR THE MULTI-CITY MODELS TRAINED ON ALL FOUR CITIES REPRESENTS THE TEST CITY, AND (II) FOR OUR ZERO-SHOT MODEL REPRESENTS THE TEST CITY. BEST RESULTS PER METRIC ARE IN **BOLD** AND SECOND-BEST ARE UNDERLINED. WITH AN ASTERISK*, WE INDICATE WHETHER THE ZERO-SHOT PREDICTION SURPASSES ALL BASELINES.

Metric (JSD)	Distance (↓)	Radius (↓)	Duration (↓)	Periodicity (↓)	I-rank (↓)	G-rank (↓)	Distance (↓)	Radius (↓)	Duration (↓)	Periodicity (↓)	I-rank (↓)	G-rank (↓)
	City A						**City B**					
RNN	0.0882	0.5130	0.0140	0.4644	0.0177	0.0036	0.0402	0.2255	0.0057	0.2210	0.0562	0.0047
LSTM	0.1477	0.4586	0.0478	0.5145	0.0177	**0.0026**	0.0483	0.1445	0.0138	0.2166	0.0523	0.0078
MoveSim [42]	0.0549	0.2670	0.2342	0.5778	0.0707	0.0063	0.0511	0.3142	0.2311	0.4652	0.0707	**0.0045**
Transformer	0.0765	0.3358	0.0206	0.2159	0.0523	0.0482	0.0135	0.0270	0.0074	0.0168	0.0523	0.0150
COLA [6]	0.0081	0.0857	0.0115	0.0667	0.0523	0.0895	0.0038	0.0502	0.0076	0.0338	0.0523	0.0261
H0xtra (ours)	0.0846	0.0770*	0.0524	0.4282	0.0177	0.0855	0.0134	0.0125*	0.0114	0.1543	0.0562	0.0261
H0xtra-SC (ours)	**0.0007**	0.0018	**0.0002**	**0.0127**	**0.0000**	0.0717	**0.0007**	**0.0035**	0.0007	**0.0052**	**0.0000**	0.0126
H0xtra-MC (ours)	0.0018	**0.0016**	0.0011	0.0356	0.0347	0.1003	0.0013	0.0038	**0.0003**	0.0194	**0.0000**	0.0410
	City C						**City D**					
RNN	0.0212	0.1570	0.0022	0.1424	0.0562	0.0189	0.1010	0.4682	0.0126	0.3446	**0.0177**	0.0384
LSTM	0.0269	0.0909	0.0062	0.1366	0.0562	0.0241	0.1274	0.3351	0.0309	0.3395	0.0523	0.0485
MoveSim [42]	0.0624	0.3715	0.6171	0.6562	0.0707	**0.0003**	0.0687	0.2165	0.1819	0.5301	0.0707	**0.0176**
Transformer	0.0119	0.0298	0.0030	**0.0048**	0.0523	0.0123	0.0704	0.2660	0.0173	0.1613	**0.0177**	0.0233
COLA [6]	0.0051	0.0510	0.0100	0.0369	0.0523	0.0392	0.0035	0.0412	0.0057	**0.0237**	0.0523	0.0623
H0xtra (ours)	0.0328	0.0425	0.0184	0.2116	0.0523	0.0303	0.0102	0.0150*	0.0048*	0.1253	0.0562	0.0564
H0xtra-SC (ours)	**0.0005**	0.0074	0.0007	0.0365	0.0347	0.0371	**0.0007**	0.0047	0.0004	0.0324	0.0347	0.0583
H0xtra-MC (ours)	0.0016	**0.0043**	**0.0005**	0.0237	**0.0000**	0.0344	0.0019	**0.0023**	**0.0003**	0.0249	0.0347	0.1281

Color coding: Single-city approaches, multi-city approaches, zero-shot cross-city approach

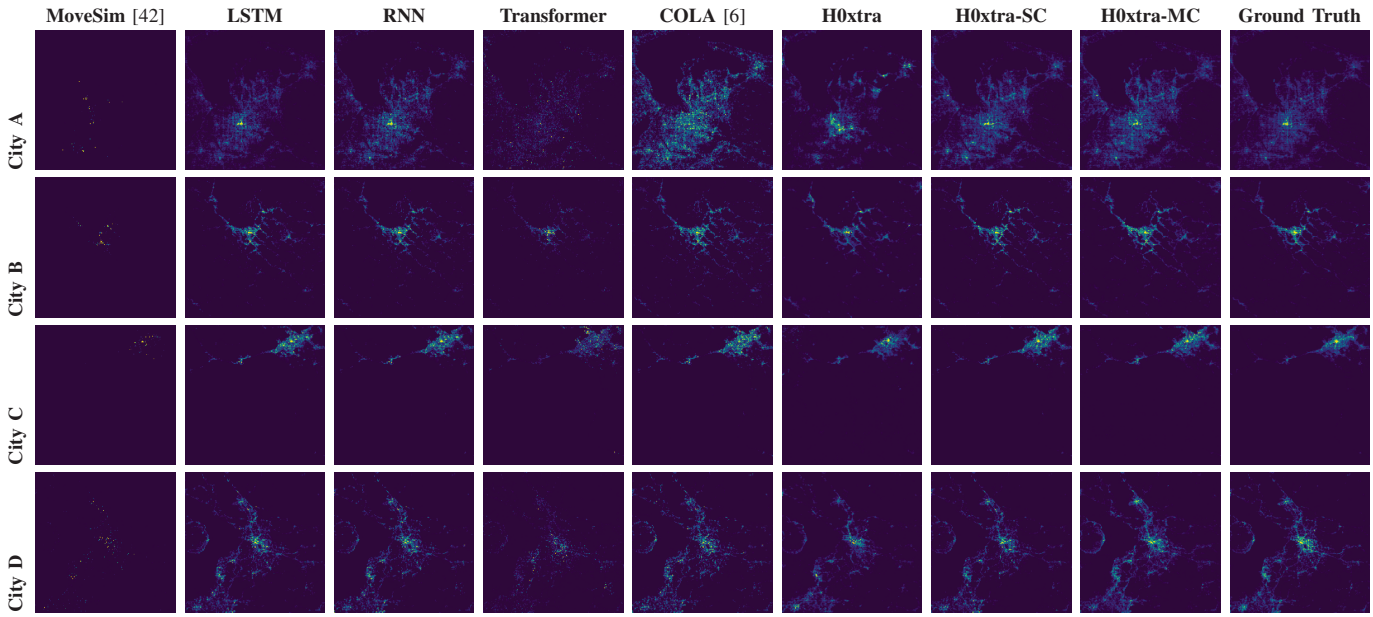


Fig. 4. Aggregated population density from generated trajectories and overall real data. Lighter colors refer to higher population volumes. To improve contrast, values above the 99.7th percentile were clipped.

1) Effect of Number of Source Cities: The source cities play a central role for the zero-shot performance, as they form the basis for learning universal trajectory representations. This section investigates how the number of source cities and their training data volume affect generalization to unseen cities.

As shown in Table IV, zero-shot performance generally improves with the number of source cities, confirming successful domain generalization.

The only exemption to this pattern is city A. Exclusively training on city A (as base city and single source city) already yields strong zero-shot results, and adding more cities improves results only marginally or even degrades performance. In contrast, using only city C, which is spatially less extensive

TABLE IV

ZERO-SHOT PERFORMANCE AND RELATIVE IMPROVEMENTS WITH AN INCREASING NUMBER OF SOURCE CITIES. THE RESULTS ARE AVERAGED (I) ACROSS ALL POSSIBLE SETS OF SOURCE CITIES THAT INCLUDE THE BASE CITY, AND (II) ACROSS ALL CORRESPONDING TARGET CITIES. THE LAST COLUMN (4 SOURCE CITIES) REFERS TO THE NON-ZERO-SHOT PERFORMANCE AVERAGED ACROSS ALL CITIES.

Base City	Metric	Number of Source Cities							
		1 (Base City)	2		3		4	
A	@1 (↑)	0.282	0.283	+0.4%	0.295	+4.2%	0.314	+6.5%
	dist. err. (↓)	2154	2177	+1.1%	2115	-2.8%	2031	-4%
B	@1 (↑)	0.201	0.242	+20.4%	0.276	+14%	0.314	+13.8%
	dist. err. (↓)	2679	2442	-8.8%	2161	-11.5%	2031	-6%
C	@1 (↑)	0.111	0.247	+122.5%	0.282	+14.2%	0.314	+11.3%
	dist. err. (↓)	20067	2477	-87.7%	2218	-10.5%	2031	-8.4%
D	@1 (↑)	0.186	0.256	+37.6%	0.290	+13.3%	0.314	+8.3%
	dist. err. (↓)	3284	2391	-27.2%	2203	-7.9%	2031	-7.8%

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and has a significantly smaller amount of training data, leads to substantially worse generalization that is significantly improved by further source cities. This performance difference cannot be explained by lower data volume alone: cities B, C and D have similar amounts of training trajectories available, yet models exclusively trained on city B or D outperform the model trained only on city C by a large margin. This finding suggests that spatial layout is a central factor in zero-shot cross-city generalization.

Finally, in most cases, we observe a gradual decrease in relative improvements, continuing up to the performance of the non-zero-shot model (i.e., H0xtra-MC). Given that the performance of H0xtra-MC can be assumed to represent an upper bound on zero-shot performance, the steady decline indicates that strong zero-shot performance can be attained with just a few source cities.

2) Choice of Hypernetwork: The hypernetwork unifies trajectory representations and hence, attains a key role in our approach. In this section, we compare different hypernetwork architectures and assess their suitability for this task.

Table V reports the performance for three different hypernetwork architectures: an MLP, a multi-layer CNN, and the default CNN U-Net; each with three input variation: POI data only, POI data with integer coordinates, and POI data with 2D sinusoidal positional encodings. Overall, the U-Net with positional encodings surpasses all other variations in terms of accuracy and ADE.

Furthermore, the results highlight that information on spatial proximity is critical to generalization, as the MLP hypernetwork, lacking any spatial context, performs poorly. In contrast, all architectures that receive information on spatial order, either through coordinate-based input augmentation or spatial arrangement of POIs, achieve notably higher performance.

Although all three architectures yield similar ADEs when provided with sinusoidal positional encodings, the accuracy achieved with the MLP is notably inferior to that of the CNN and U-Net. While the MLP infers location embeddings from isolated POI profiles, the CNN and U-Net incorporate information from the locations' neighborhoods. Consequently, the MLP cannot learn to encode a location's role within its broader urban context. The difference in accuracy, however, demonstrates the relevance of this urban context to learning meaningful location representations.

TABLE V
ANALYSIS OF DIFFERENT HYPERNETWORKS. FOR THIS EXPERIMENT, CITIES A, B, AND C WERE USED AS SOURCE CITIES AND CITY D AS TARGET CITY. HERE, *coords* REFERS TO INTEGER COORDINATES AND *pos. enc.* TO SINUSOIDAL POSITIONAL ENCODINGS THAT WE ADDED TO EACH LOCATION'S POI VECTOR.

Architecture	@1 (↑)	@10 (↑)	MRR (↑)	ADE (↓)
MLP	0.003	0.027	0.014	38973
MLP + coords	0.057	0.325	0.142	3146
MLP + pos. enc.	0.176	0.555	0.300	2332
CNN	0.269	0.633	0.396	2708
CNN + coords.	0.270	0.636	0.398	2134
CNN + pos. enc.	0.273	0.642	0.403	2154
CNN U-Net	0.247	0.595	0.370	2706
CNN U-Net + coords	0.267	0.625	0.394	2116
CNN U-Net + pos. enc.	0.273	0.649	0.406	2088

VI. CONCLUSION

Large-scale human mobility data is rarely publicly available, hindering critical applications such as urban planning or epidemic control. In this work, we proposed H0xtra, a novel hypernetwork-based approach for zero-shot cross-city trajectory prediction that generates spatially consistent synthetic trajectories in unseen cities. For cities that completely lack mobility data, H0xtra overcomes this limitation by enabling realistic trajectory simulation using widely available POI data only.

Our results demonstrate that hypernetworks are effective for unifying trajectories across cities in a shared representation space, laying the groundwork for a new paradigm in multi-city trajectory modeling. Our findings further confirm that universal trajectory patterns exist across cities and can be largely inferred from POI distributions and urban structure.

Future work could explore extensions using graph neural networks to better capture inter-city dependencies or incorporate situational factors to simulate human mobility under specific scenarios.

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